

N-Body Gravitational Simulation : Effects of Numerical Integration on Energy Conservation and Orbital Stability

This paper presents an N-body gravitational simulation of the solar system and compares two numerical integration methods : Euler and Leapfrog, in terms of energy conservation and orbital stability. Total system energy is tracked over one Mercury orbital period as a stability metric. Results show that Leapfrog integration produces approximately 10,000 times less energy drift than Euler, confirming that symplectic integrators are significantly more suitable for long term gravitational simulations.

0. Introduction :

Gravitational systems describe the motion of bodies interacting through Newton's law of universal gravitation. While simple two-body systems can be solved analytically, systems with multiple interacting bodies quickly become complex which forms nonlinear dynamical systems. Because of this complexity, Numerical simulation is required to approximate their behavior over time. By representing time into small steps. It becomes possible to continually compute the motion of each body under the influence of gravitational forces. The goal of this project is to develop an N-body gravitational simulation and investigate how different numerical integration methods affect energy conservation and long term orbital stability.

1. Physical Model :

The system is modeled using Newton's law of universal gravitation, where each body exerts an attractive force on every other body. The magnitude of this force is given by:

$$F = G \frac{m_1 m_2}{r^2}$$

Where r is the distance between the two planets.

Since force is a vector quantity, it is decomposed into components along the x and y axes based on the positions of the bodies.

Using Newton's second law. the acceleration of each body is determined by :

$$\vec{a} = \frac{\vec{F}}{m}$$

Each body in the system is described by its position and velocity. Velocity is defined as the rate of change of position :

$$\vec{v} = \frac{d\vec{x}}{dt}$$

Similarly, acceleration is defined as the rate of change of velocity :

$$\vec{a} = \frac{d\vec{v}}{dt}$$

Together, these relationships build the connection between force, acceleration, velocity, and position, forming the basis for describing the motion of the system.

2. Numerical Methods :

2.1 Euler method :

Physical motion is continuous, while computers operate in discrete time. To simulate motion numerically, time is divided into small increments denoted by dt . The system is then updated step by step, approximating continuous evolution through discrete updates.

At each timestep, the velocity and position of each body are updated based on the current acceleration. The size of dt plays a critical role in the accuracy of the simulation. Smaller values of dt improve accuracy and stability but increase computational cost, while larger values lead to faster simulations at the expense of accuracy.

Euler integration is a simple method used to approximate the solution of differential equations of motion. In this method, the velocity is updated using the current acceleration, and the position is then updated using the new velocity :

$$v_{new} = v + a \cdot dt$$

$$x_{new} = x + v \cdot dt$$

This method effectively approximates motion as a sequence of steps, assuming that acceleration remains constant during each time step.

However, Euler integration has significant limitations. It does not conserve energy, and small numerical errors introduced at each timestep accumulate over time. As a result, the system may either gain or lose energy, leading to unstable orbits and deviation from the expected physical behavior. These errors become more pronounced when larger timesteps are used.

2.2 Leapfrog Method :

Euler integration introduces numerical errors that accumulate over time. leading to gain or loss of energy. This makes it unsuitable for long-term simulations of orbital systems.

To avoid this, more stable integration methods are used. One such method is Leapfrog integration, which is specifically designed to improve long term stability in simulations.

Leapfrog integration updates velocity and position at different times. Velocity is computed at half timesteps :

$$v_x^{t+\frac{dt}{2}} = v_x^t + a_x^t \cdot \frac{dt}{2}$$

$$v_y^{t+\frac{dt}{2}} = v_y^t + a_y^t \cdot \frac{dt}{2}$$

Position is computed at full timesteps. Which makes the position and velocity continuously update each other :

$$x^{t+dt} = x^t + v_x^{t+\frac{dt}{2}} \cdot dt$$

$$y^{t+dt} = y^t + v_y^{t+\frac{dt}{2}} \cdot dt$$

Then it recomputes acceleration at new position, where the new accelerations are recomputed from the updated positions using Newton's law of gravitation :

$$a_x^{t+dt}, \quad a_y^{t+dt}$$

After that it recompute the final velocity :

$$v_x^{t+dt} = v_x^{t+\frac{dt}{2}} + a_x^{t+dt} \cdot \frac{dt}{2}$$

$$v_y^{t+dt} = v_y^{t+\frac{dt}{2}} + a_y^{t+dt} \cdot \frac{dt}{2}$$

Unlike Euler, Leapfrog better approximates continuous motion because it distributes error symmetrically and it does not always push the system in one direction

Leapfrog is a symplectic integrator, meaning it preserves the geometric structure of the system. As a result of that the total energy does not drift indefinitely but instead.s it oscillates around the true value and orbits remain stable over long simulations. But, it can be more complicated than Euler. It is worth noting that Leapfrog still introduces small numerical errors, and its accuracy remains dependent on the choice of timestep dt .

3. Results :

3.1 Energy Evolution – Euler :

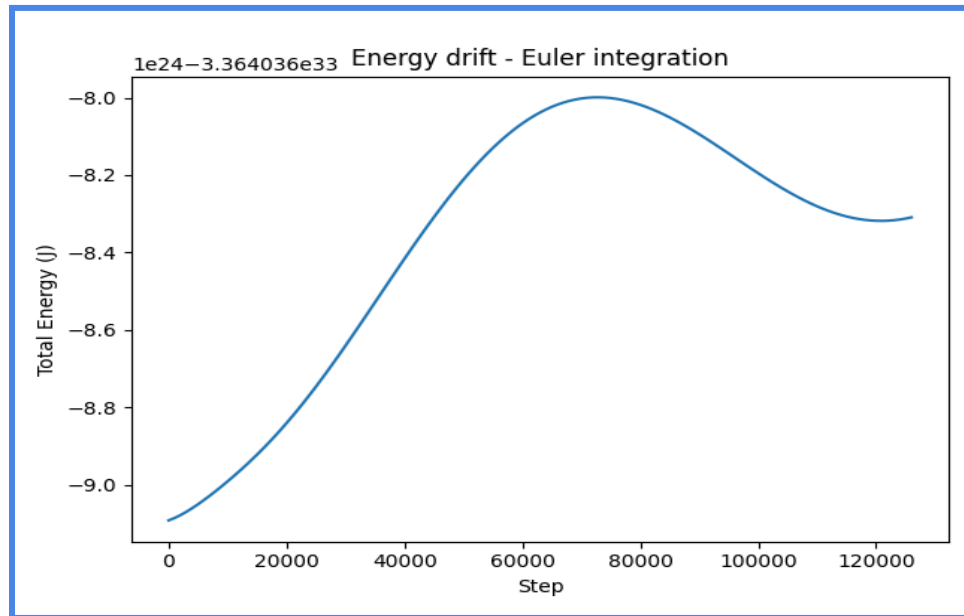


Figure 1 : Energy drift for Euler integration over one Mercury orbital period.

Observing the energy of the system over time. The energy does not stay the same. And errors start to accumulate over time steps which leads to instability and inaccuracy.

3.2 Energy Evolution – Leapfrog :

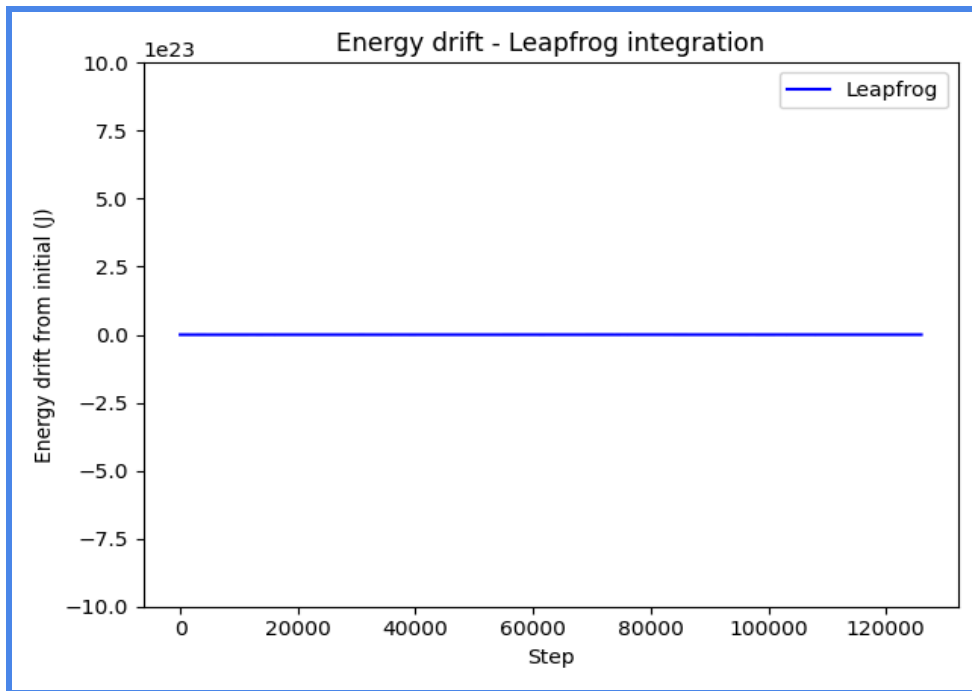


Figure 2 : Energy drift for Leapfrog integration over one Mercury orbital period.

Observing the energy of the system over time. Energy oscillates and remains bounded over large timesteps.

3.3 Method Comparison :

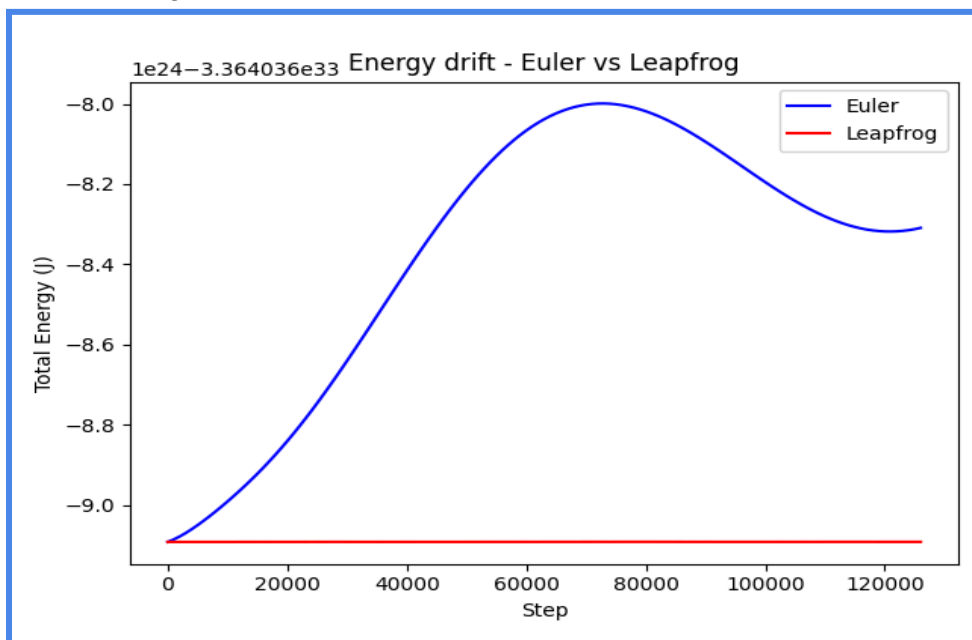


Figure 3 : Energy drift for Euler and Leapfrog integrations over one Mercury orbital period.

Observing the energy of the system over time with the two methods. It shows that Leapfrog integration remains stable, consistent, and bounded over large timesteps, while Euler integration shows a significant change in energy over the same number of steps. Note that the two graphs (figure 1 and figure 2) use different y axis scales. Leapfrog drift is approximately 10,000 times smaller than Euler's.

4. Analysis :

4.1 Energy Drift in Euler :

The Euler method fails with huge time steps primarily due to numerical instability, where the solution diverges or oscillates wildly. And because Euler updates are one sided, errors accumulate in the same direction. This causes continuous energy gain or loss in energy and unstable orbits.

4.2 Stability of Leapfrog :

For Leapfrog, updates are symmetric. Errors cancel out over time, which prevents the energy from drifting. Instead the energy oscillates around a constant value. This makes the errors balance instead of accumulating.

4.3 Role of Time Step (dt):

Larger dt values lead to significant errors and instability, while smaller dt values give better approximation and more realistic motion. This tradeoff applies to both methods, but Leapfrog remains far more stable than Euler at the same timestep.

4.4 Physical Meaning :

Isolated gravitational systems conserve total energy. The motion of planets is controlled by this conservation, which ensures that orbits remain stable over long periods of time. A numerical simulation should therefore reproduce this fundamental property. However, the results show that not all integration methods respect this constraint equally.

Euler integration introduces systematic numerical errors. Because its updates are not symmetric, small errors in energy accumulate over time. This leads to a continuous drift in total energy, causing orbits to decay or expand. As a result, the simulated system deviates from physically realistic behavior. In contrast, Leapfrog integration preserves the qualitative structure of the system. Although it does not maintain perfectly constant energy, it confines the error within bounded oscillations. This behavior reflects the underlying conservation laws more accurately, allowing the system to maintain stable orbits over long durations. This demonstrates that the choice of numerical method is not only a matter of computational accuracy, but also of physical loyalty. A method that respects conservation laws produces simulations that remain consistent with real world dynamics, while a method that violates them leads to fundamentally incorrect behavior over time.

5. Conclusion :

This study implemented an N-body gravitational simulation and compared Euler and Leapfrog integration methods in terms of energy conservation. The results show that Euler integration produces a significant energy drift of approximately 1.1×10^{24} J, over one Mercury orbital period, while Leapfrog integration drift remains bounded at approximately 1.3×10^{20} J, roughly 10,000 times smaller. This confirms that the choice of numerical integrator has a fundamental impact on the physical accuracy of the simulation. Leapfrog integrator, is the correct choice for long term orbital simulations where energy conservation is required.